



EXAM SPOTLIGHT

Certyfikat PMP: Quality Control Techniques

Mastering the tools, metrics, and processes to ensure project excellence.

dr Zbigniew Galar

Bridging the Plan and the Product

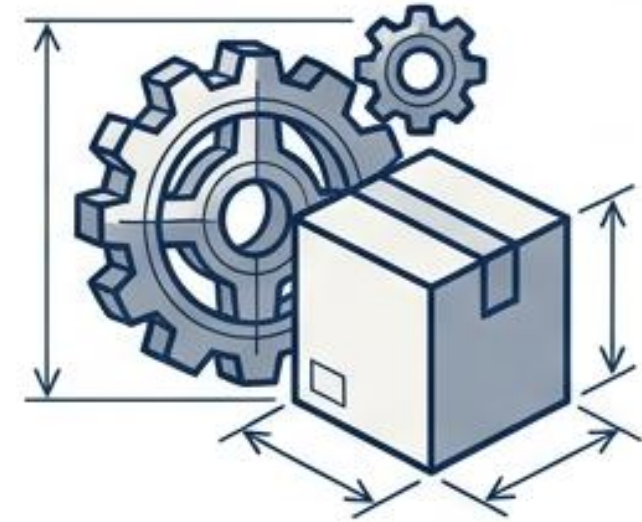


The Plan

(Management Performance:
Schedule/Cost)



Control
Quality

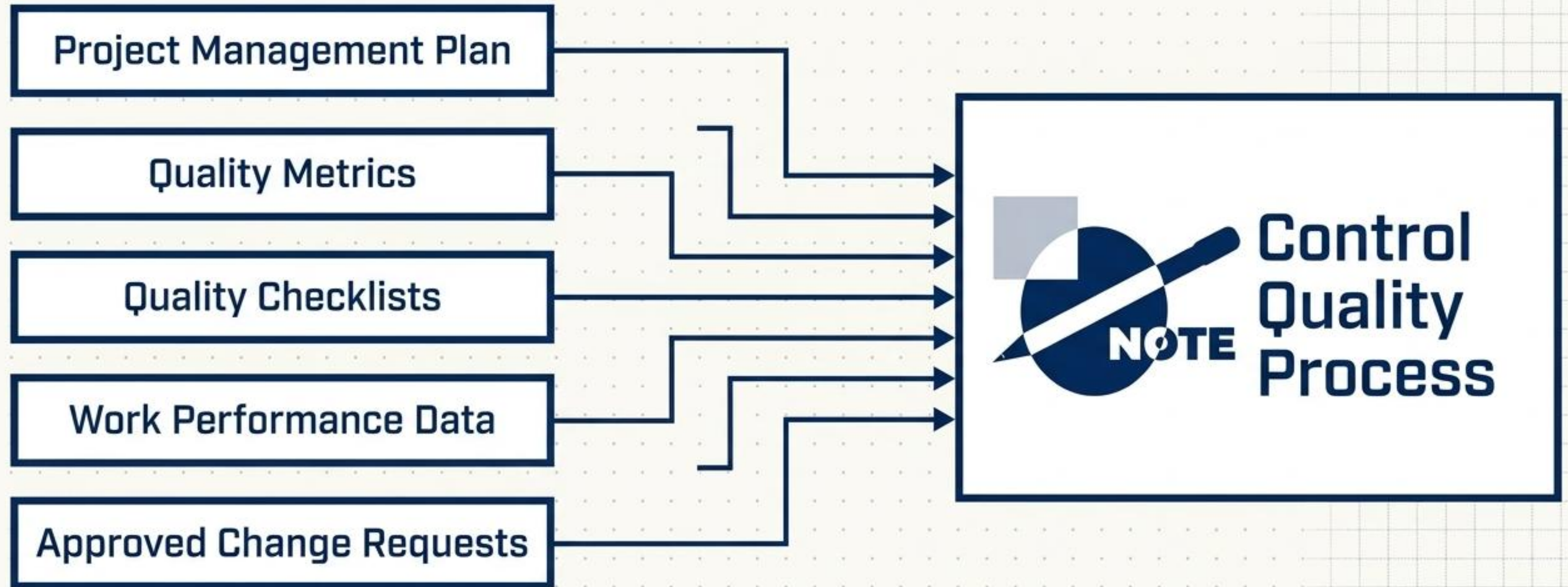


The Product

(Conformance to
Requirements)

Key Concept: Control Quality monitors work results to verify they comply with the standards set in the Quality Management Plan.
Goal: Continuously identify and remove the root causes of unacceptable results.

What Feeds the Control Quality Process?



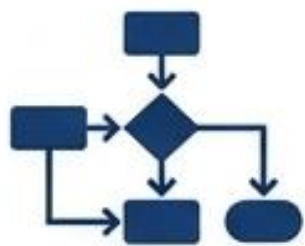
You cannot control quality without a predefined baseline and real-world performance data to measure against.

The PM's Quality Toolkit: The 7 Basic Tools

Finding Root Causes



Cause-and-Effect
(Ishikawa/Fishbone) Diagrams

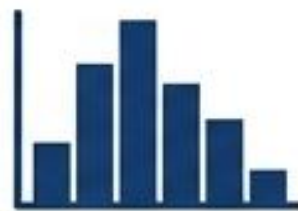


Flowcharts

Tracking & Measuring



Control Charts

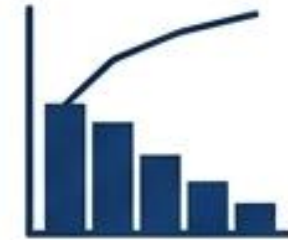


Histograms



Checksheets

Prioritizing & Correlating



Pareto Diagrams



Scatter Diagrams

These tools fuel the Plan-Do-Check-Act cycle to identify defects and ensure conformity.

Anatomy of a Control Chart: Monitoring Stability



Exam Spotlight:

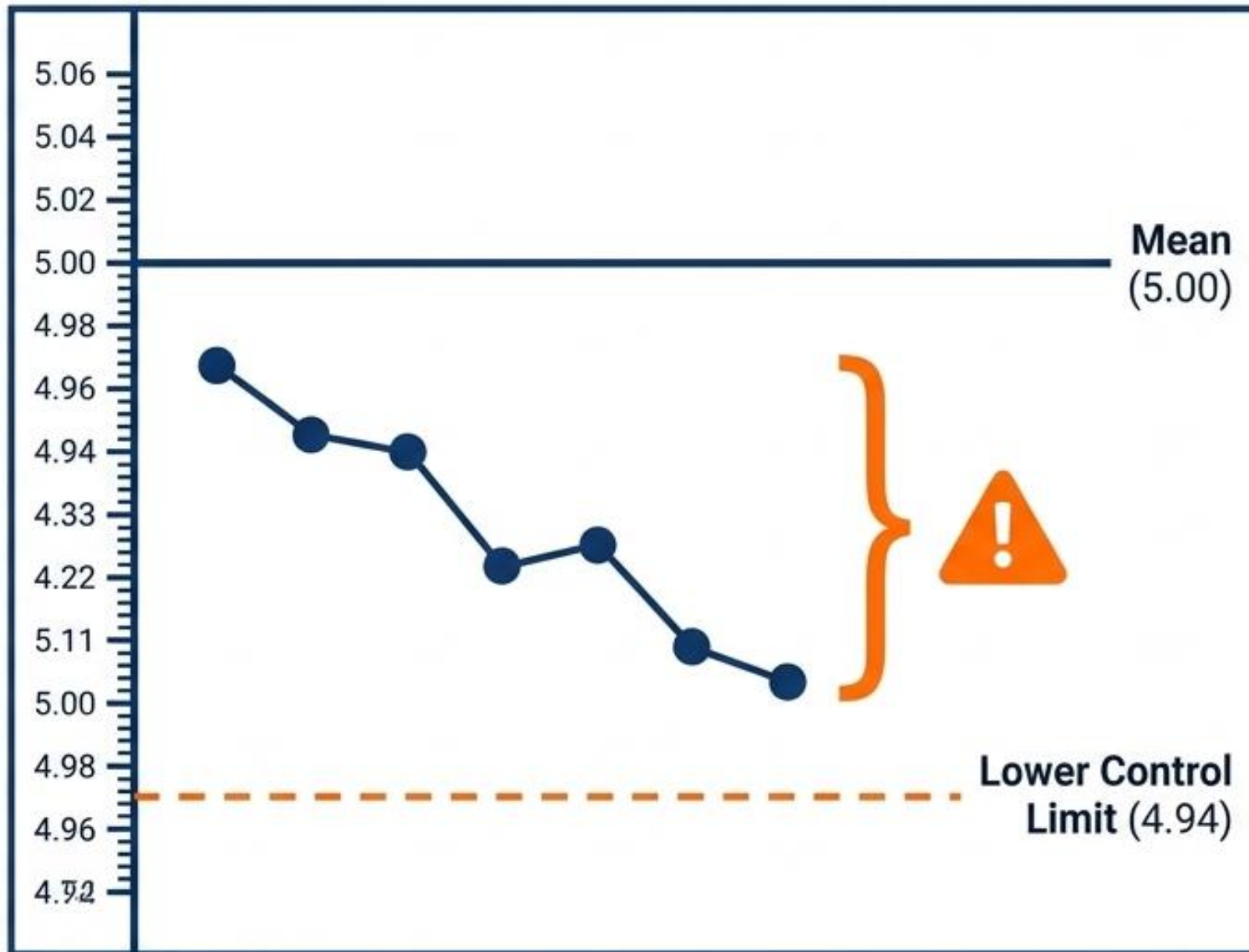
If a process is in control, it should NOT be adjusted!

Variance Diagnostics: Why Do Differences Occur?

The Exception	
Special-Cause Variance	Not common to the process. Caused by isolated anomalies (e.g., a missed procedure, a suddenly broken machine requiring calibration).
The System	
Common-Cause Variance <i>*Note: Requires management approval to reorganize the process.</i>	● Random: Normal variations occurring unpredictably. ● Known/Predictable: Unique to specific product characteristics. ● Always Present: Inherent system variability (human error, environment).



Exam Spotlight: The Rule of 7 Heuristic



The Concept:

If 7 consecutive points fall on one side of the mean, the process may be out of control.

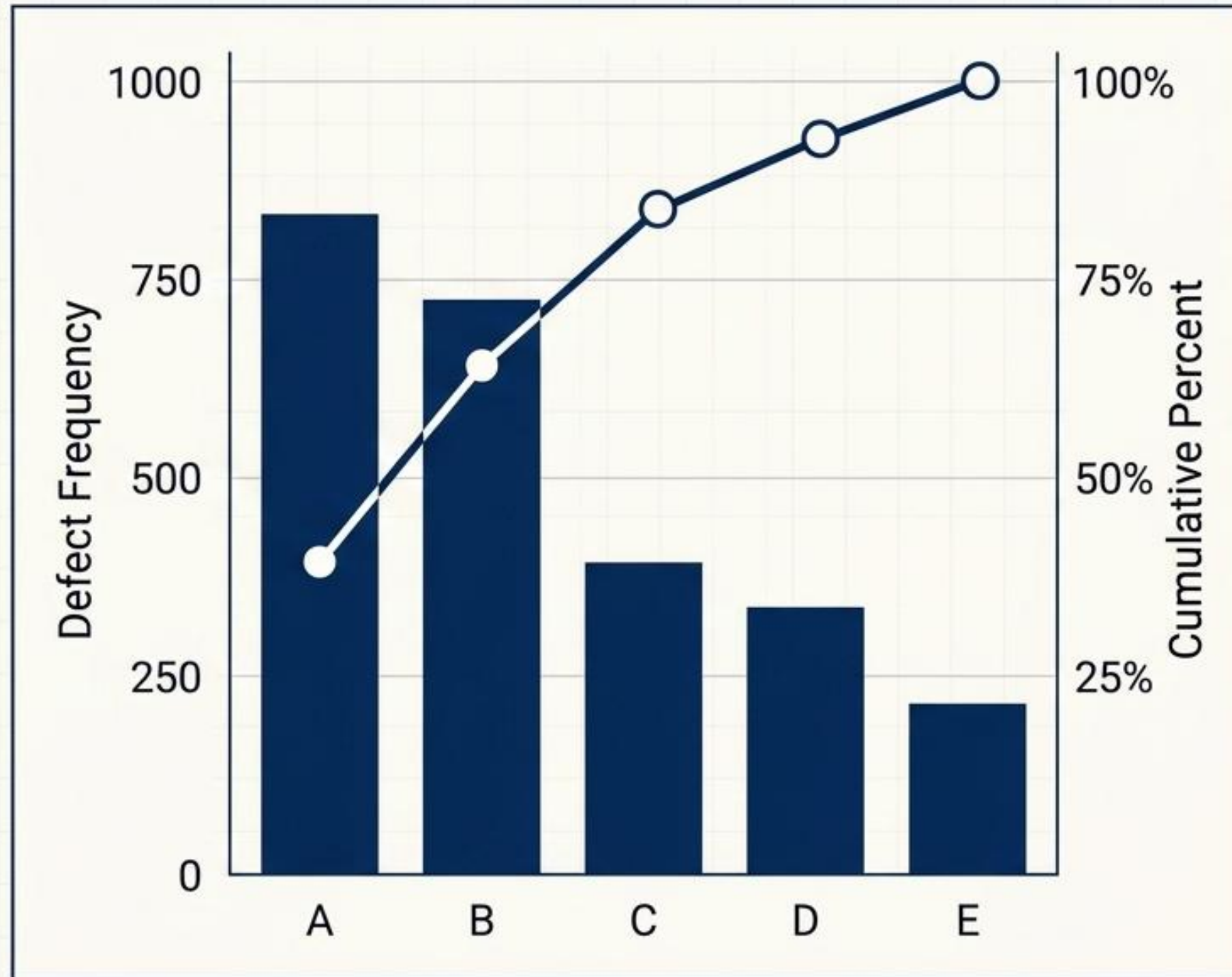
The Caveat:

This holds true even if no single point breaches the upper or lower control limits.

Action:

This pattern indicates **non-random factors are secretly influencing** the results. Immediate investigation is required.

Pareto Diagrams: Prioritizing the Pain



Concept:

Derived by Vilfredo Pareto. In quality management, the 80/20 rule dictates that a small number of causes (20%) create the majority of the problems (80%).

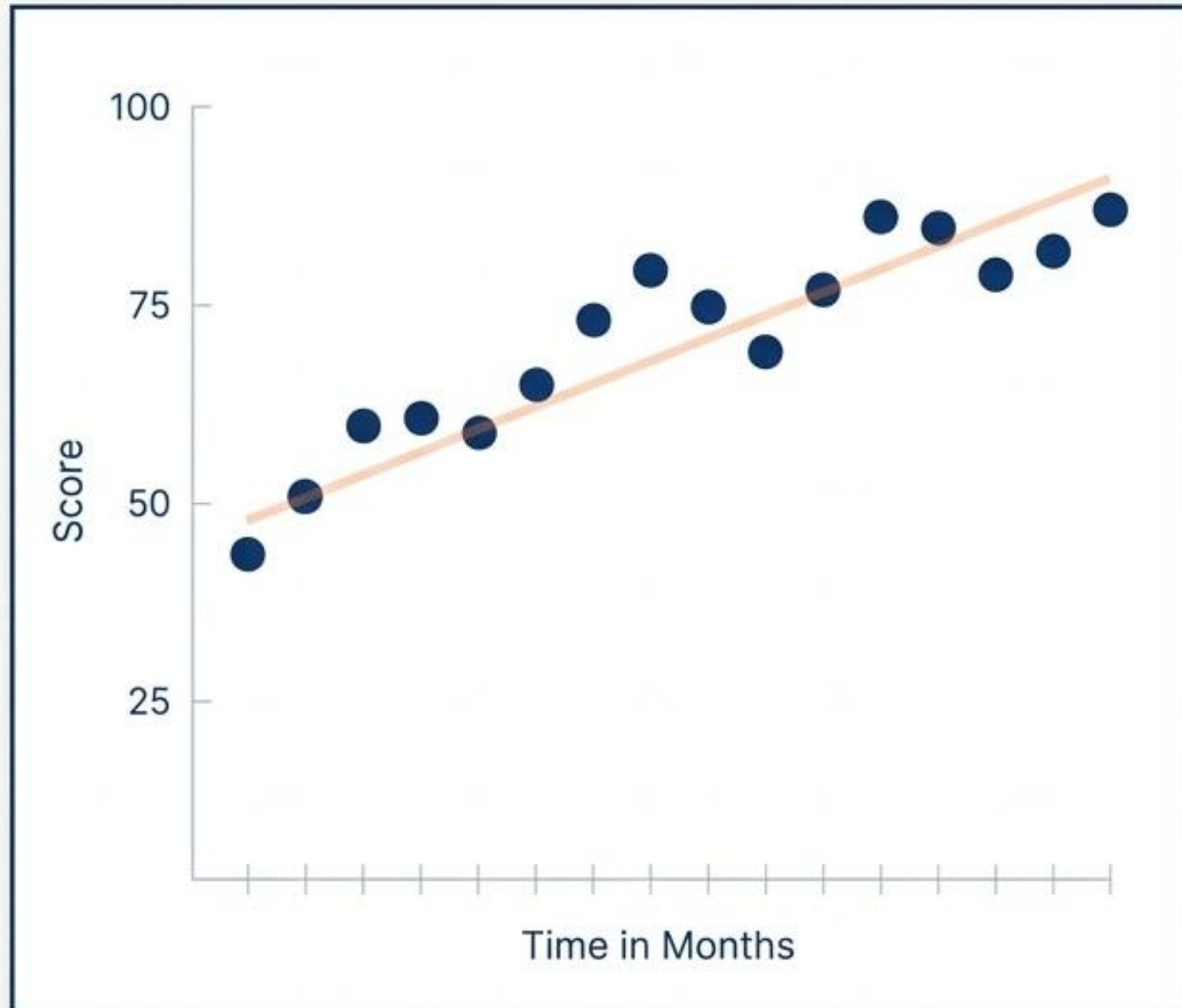
How to Read:

Rank-orders defects by their frequency.

Strategic Takeaway:

Spend the majority of your time fixing the few systemic problems on the far left of the chart to yield the highest overall benefit.

Scatter Diagrams: Proving Relationships

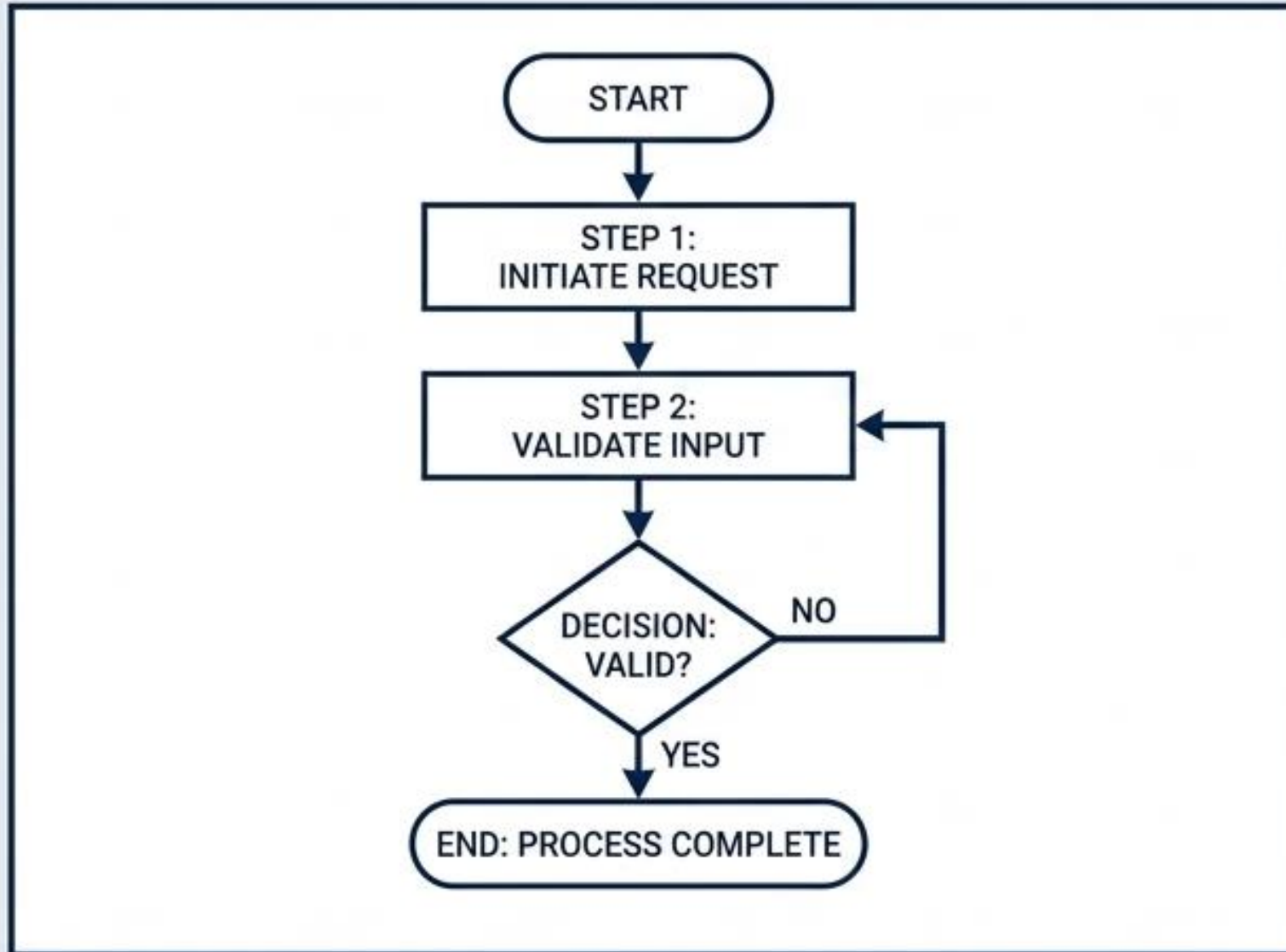


Concept: Plots two variables to analyze cause-and-effect relationships (e.g., does more experience lead to higher accuracy?).

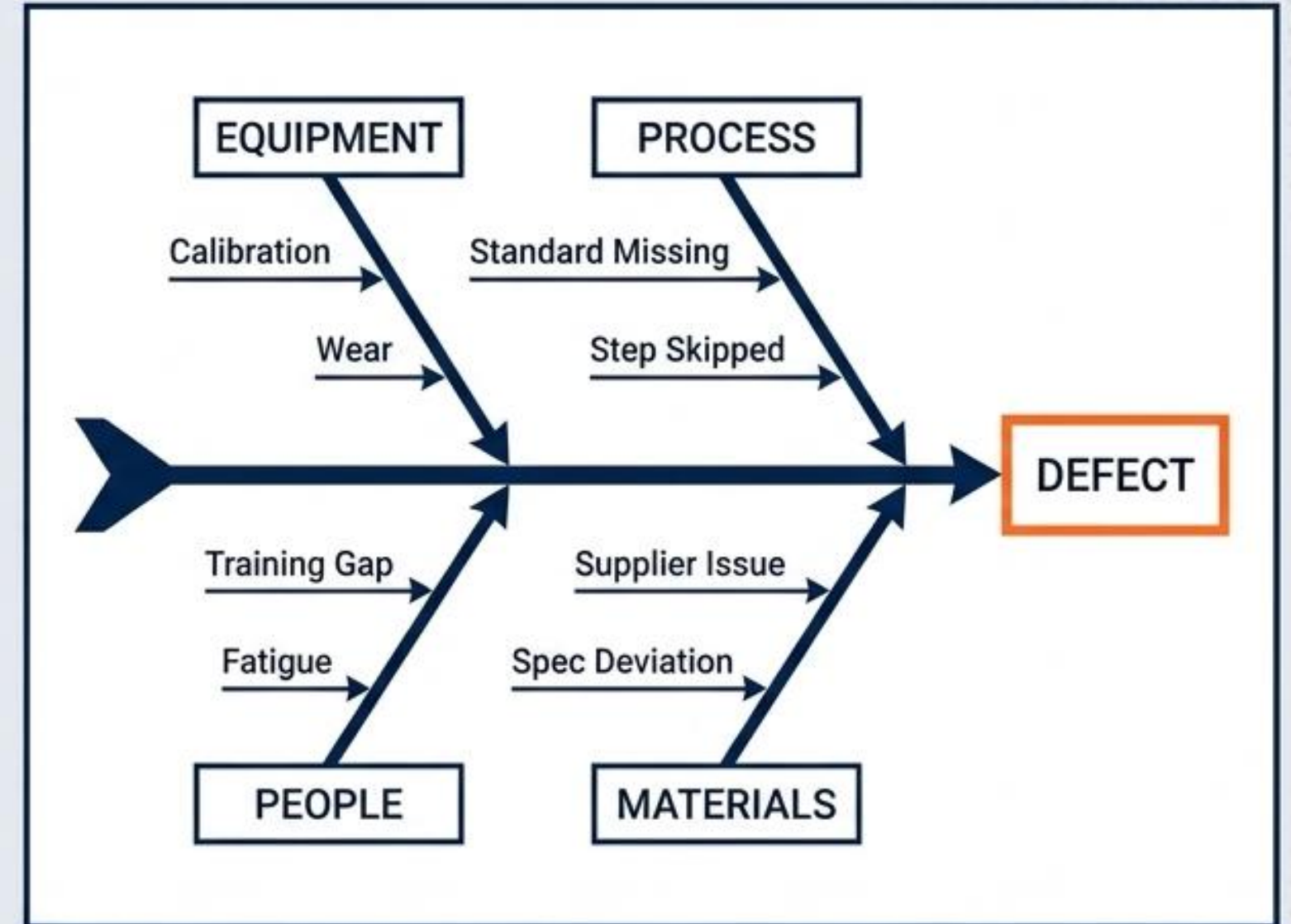
Key Rule: The closer the plotted points resemble a diagonal line, the stronger the relationship between the independent input and the dependent output.

Use Case: Validating root causes before implementing a process fix.

Process Mapping: Visualizing the Flow

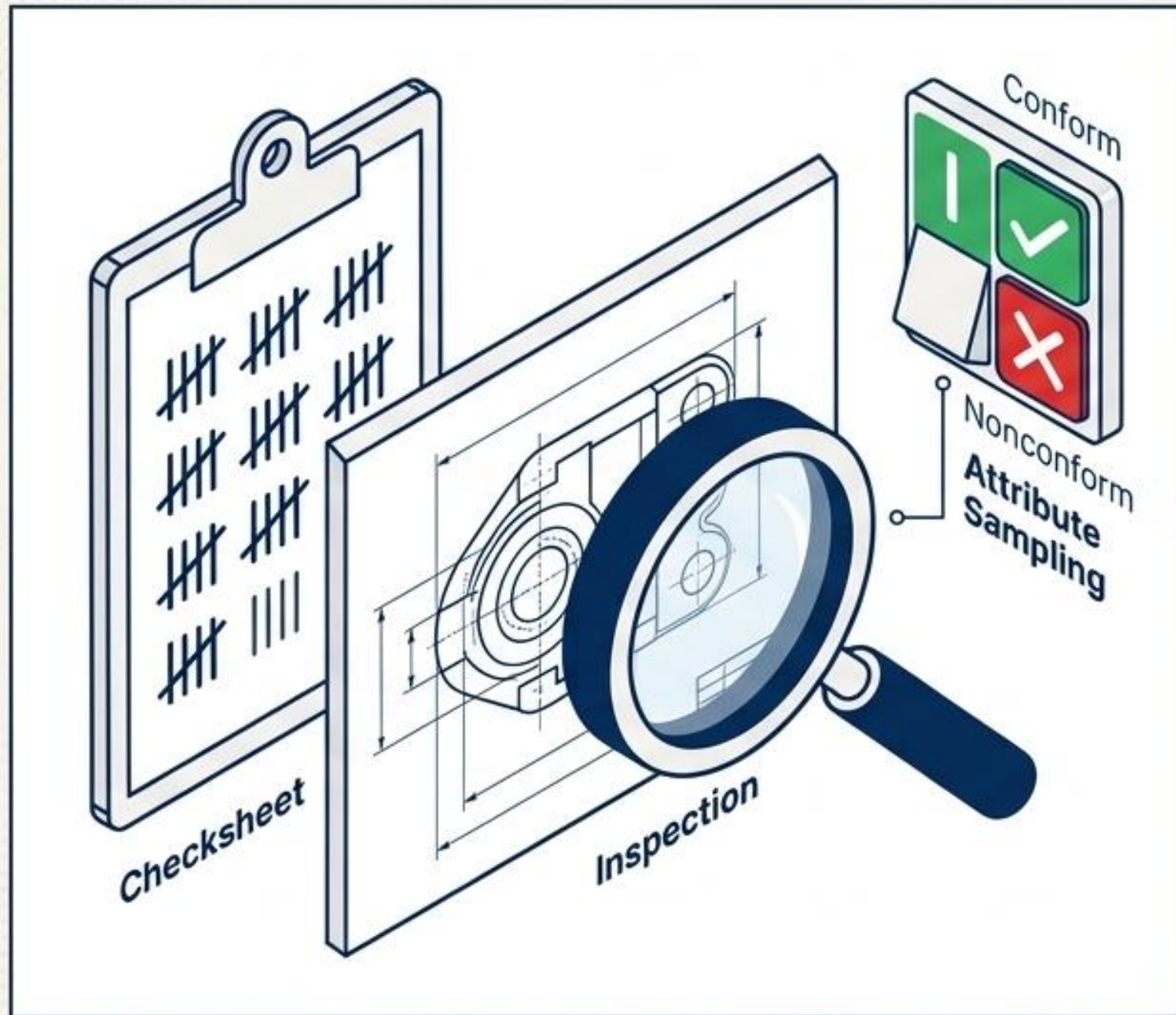


Flowcharts (Anticipating Problems): Show the logical steps to accomplish an objective. Helps the team anticipate where quality bottlenecks might occur to develop workarounds.



Ishikawa / Fishbone (Finding Problems): Works backward from an identified defect to categorize and trace its absolute root cause.

The Mechanics of Checking: Inspections & Checksheets

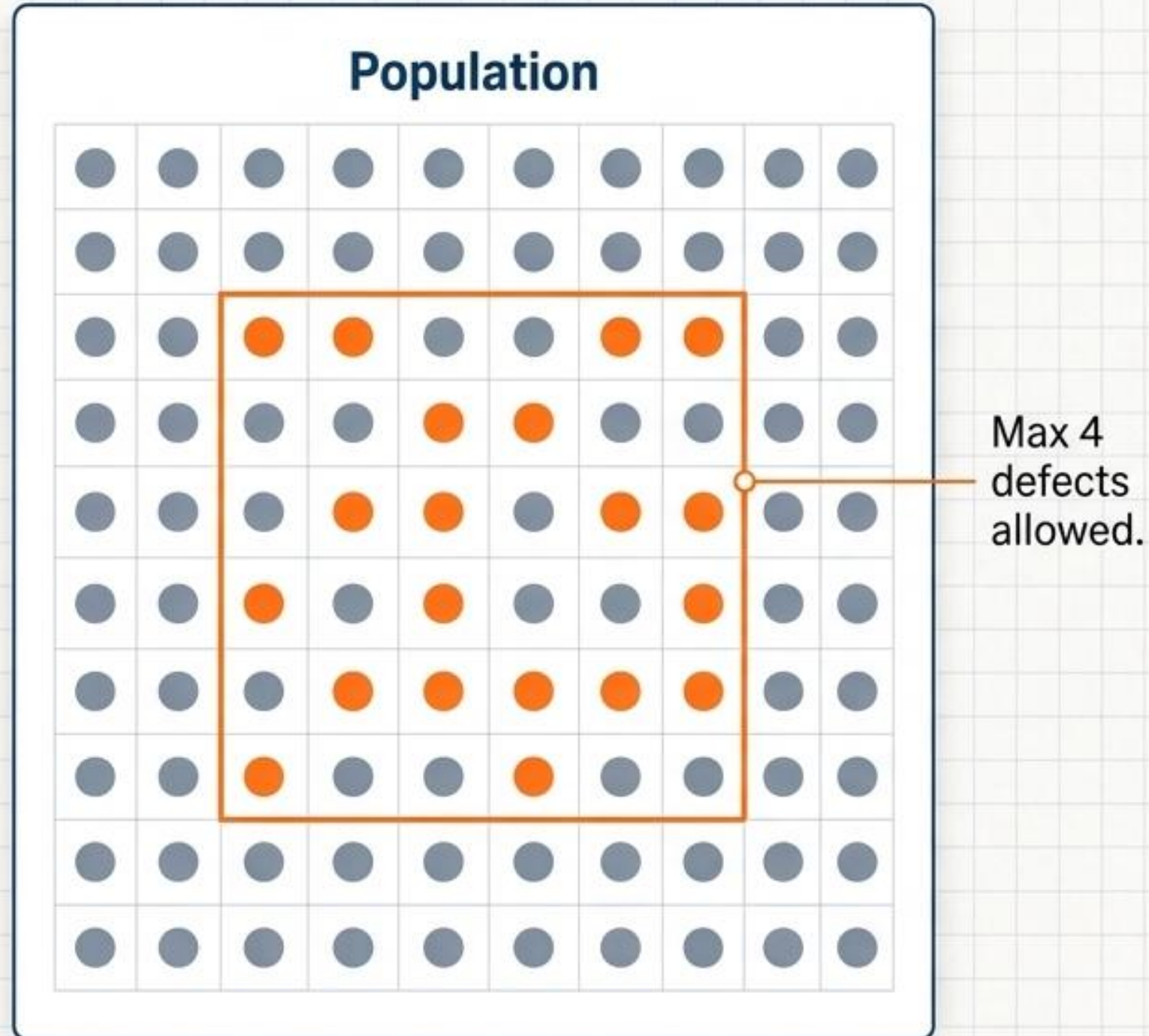


Checksheets (Tally Sheets): Used to actively organize defect frequency data during an inspection. This is the raw data that feeds into Pareto diagrams.

Inspection (Audits/Walkthroughs): Physically measuring or testing results against exact tolerances (e.g., 5.00 mm +/- 0.02 mm).

Attribute Sampling: A strict binary inspection. The result either conforms or non-conforms. A simple pass/fail decision.

Efficiency in Measurement: Statistical Sampling



The Challenge: Inspecting every individual component is often too time-consuming and cost-prohibitive.

The Solution: Test a statistically significant sample to determine if the entire population falls within acceptable variances.

The Standard:

- The Quality Management Plan dictates the required rigor.
- ± 2 Standard Deviations = 95.44% of population.
- ± 3 Standard Deviations = 99.73% of population.

EXAM SPOTLIGHT: Prevention vs. Inspection



Prevention (Process)

Keeps errors from occurring in the process itself. Always cheaper.

Low Cost / High Value



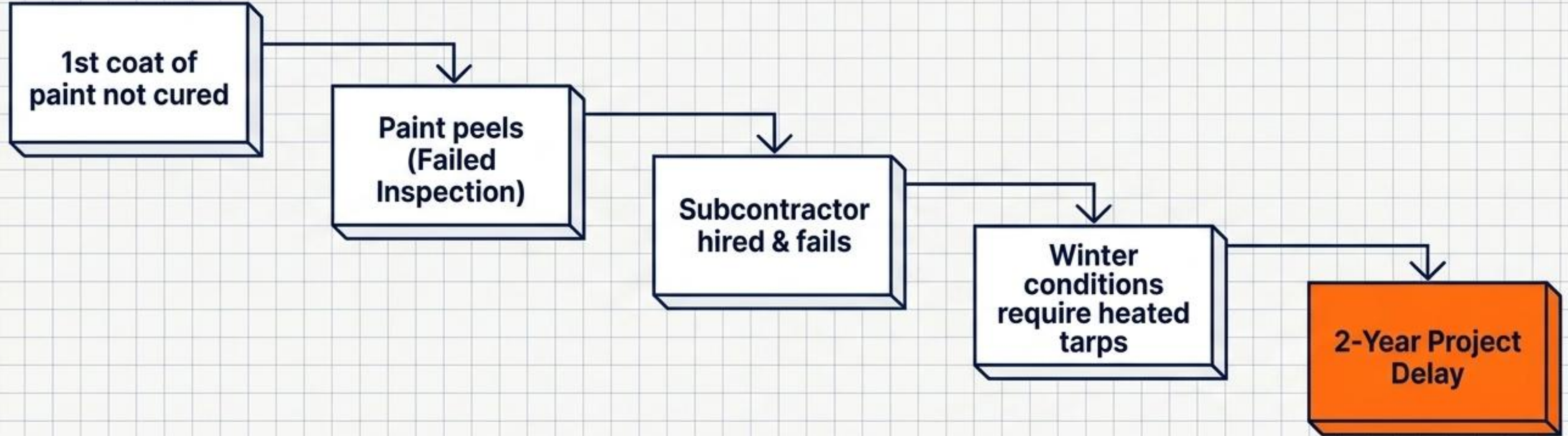
Inspection (Product)

Keeps already-made errors from reaching the customer. Extremely costly.

High Cost (Rework, Labor, Waste)

The Theory of Zero Defects: Pioneered by Philip Crosby. Quality means “doing it right the first time.” Investing in prevention is always cheaper than paying for the cure.

The Cost of Poor Quality: The Bridge Project



Scenario:

A city bridge requires 3 coats of guaranteed 25-year paint under a fixed-price contract.

The Failure:

The contractor applies the 2nd coat before the 1st coat cures, violating the quality standard.

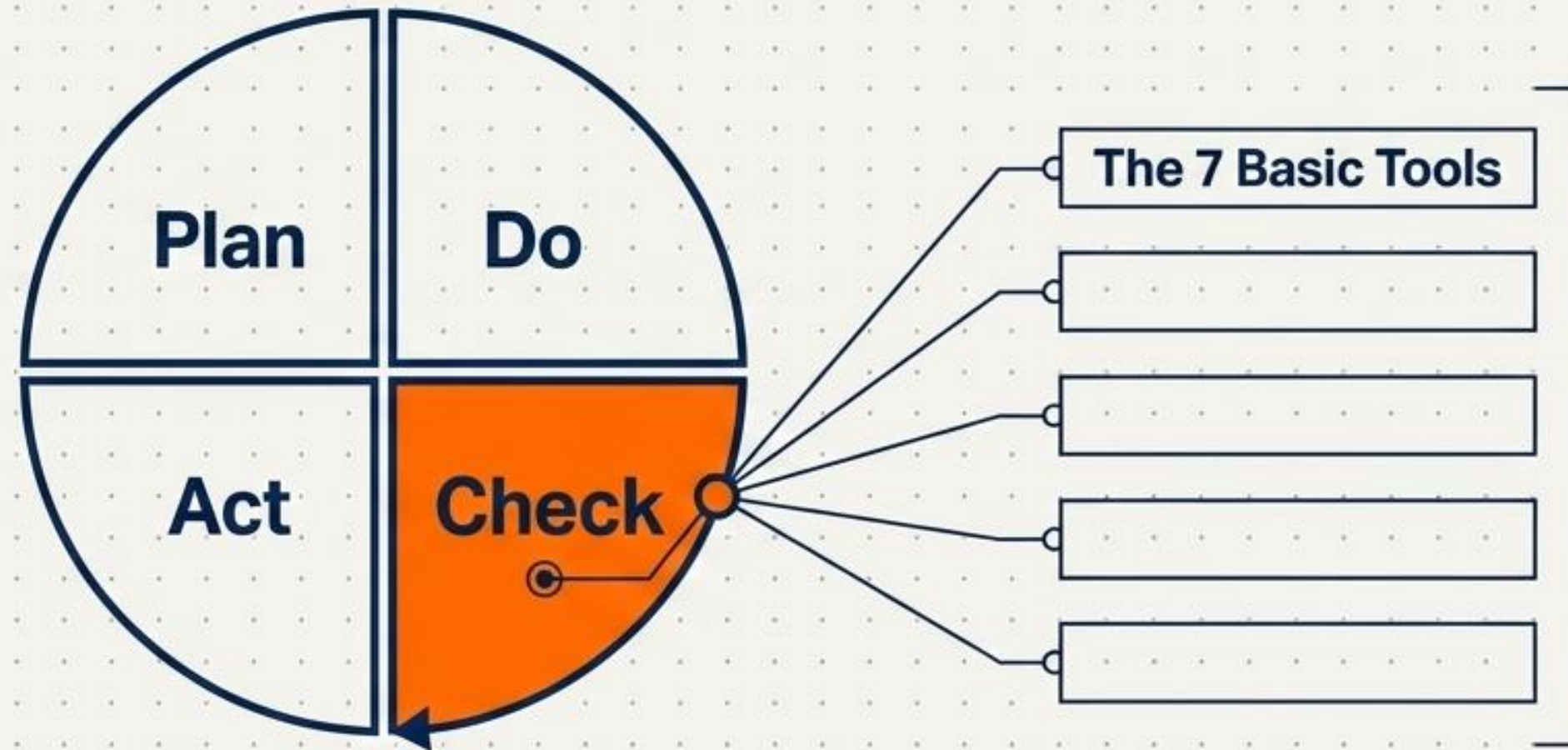
The Fallout:

Failed inspection leads to subcontracting, missed deadlines, weather complications, and massive rework costs.

The Lesson:

The fixed-price contractor's profit was entirely destroyed by poor quality. "An ounce of prevention is worth several gallons of cure."

Closing the Loop: Quality is a Continuous Cycle



- **Control Quality** represents the critical '**Check**' phase of the project lifecycle.
- By tracking variance (Control Charts), diagnosing root causes (Ishikawa), and committing to **Zero Defects**, Project Managers do more than fix broken parts.
- They **protect** the schedule, secure the budget, and guarantee excellence for the client.